Mathlearn · 6-sinf · Matematika

Murakkab shakllarning yuzi va perimetri

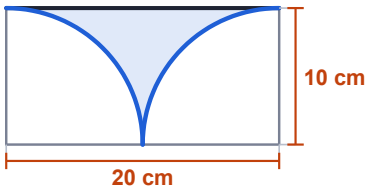
2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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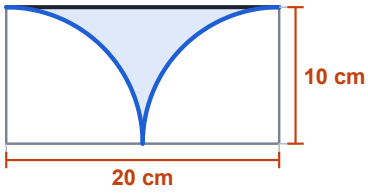
1. **To'g'ri to'rtburchak 20 cm × 10 cm. Uning pastki ikki burchagidan radiusi 10 cm bo'lgan chorak doiralar chizildi (rasm). Bo'yalgan soha yuzini toping (π = 3,14).**



A) 157 cm²
B) 114 cm²
C) 43 cm²
D) 121,5 cm²

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2. **To'g'ri to'rtburchak 20 cm × 10 cm. Uning pastki ikki burchagidan radiusi 10 cm bo'lgan chorak doiralar chizildi (rasm). Bo'yalgan soha perimetrini toping (π = 3,14).**



A) 82,8 cm
B) 51,4 cm
C) 35,7 cm
D) 71,4 cm

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3. **Konki maydoni to'g'ri to'rtburchak (50 m × 20 m) va uning ikki qisqa tomoniga yopishgan yarim doiralardan iborat. Maydon chetiga tashqaridan eni 1 m bo'lgan rezina qoplama yotqizildi. Qoplama yuzini toping (π = 3,14).**

A) 65,94 m²
B) 165,94 m²
C) 100 m²
D) 231,88 m²

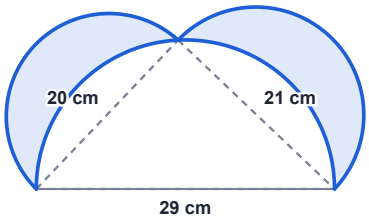
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4. **Uchburchakning tomonlari 15 m, 20 m va 25 m. Uning har bir tomoniga tashqaridan yarim doira chizildi. Hosil bo'lgan shakl (uchburchak + 3 ta yarim doira) perimetrini toping (π = 3,14).**

A) 154,2 m
B) 188,4 m
C) 94,2 m
D) 47,1 m

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5. **Naqsh to'g'ri burchakli uchburchakdan boshlab chizildi: to'g'ri burchak yonidagi tomonlari 20 cm va 21 cm, eng uzun tomoni 29 cm. Har bir tomon diametr bo'ladigan qilib yarim doiralar chizildi (rasm). Ikkita «yarim oy» — bo'yalgan qismlarning umumiy yuzini toping (π = 3,14).**



A) 210 cm²
B) 330,0925 cm²
C) 540,0925 cm²
D) 120,0925 cm²

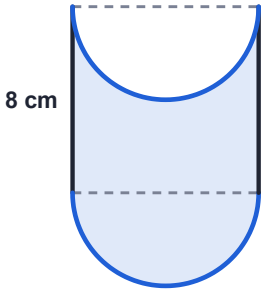
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6. **Usta tomoni 8 cm bo'lgan kvadrat koshinga naqsh chizdi: sirkul ignasini avval koshinning bir burchagiga, keyin qarama-qarshi burchagiga qo'yib, radiusi 8 cm bo'lgan ikkita yoy chizdi. Yoylar orasidagi yaproq naqsh yuzini π orqali ifodalang.**

A) (32π − 64) cm²
B) (32π − 32) cm²
C) (16π − 64) cm²
D) (64 − 16π) cm²

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7. **Kvadratning tomoni 8 cm. Tepa tomonidan yarim doira o'yib olinib, pastki tomoniga yopishtirildi (rasm). Shakl perimetrini toping (π = 3,14).**



A) 41,12 cm **B)** 32 cm
C) 57,12 cm **D)** 66,24 cm

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8. **Kvadratning tomoni 8 cm. Tepa tomonidan diametri 8 cm bo'lgan yarim doira o'yib olinib, pastki tomoniga yopishtirildi. Hosil bo'lgan shakl yuzini toping (π = 3,14).**

A) 64 cm² **B)** 89,12 cm²
C) 114,24 cm² **D)** 13,76 cm²

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9. **Doira shaklidagi gulzor markazdan chiqqan yo'lakchalar bilan 8 ta teng bo'lakka ajratilgan. Gulzorning radiusi 16 m. Bitta bo'lakni to'liq o'rab chiqish uchun necha metr to'siq kerak? (π = 3,14)**

A) 60,56 m **B)** 12,56 m **C)** 28,56 m **D)** 44,56 m

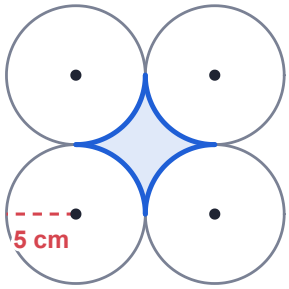
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10. **Diametri 32 m bo'lgan doira shaklidagi gulzor 8 ta teng bo'lakka ajratilgan. Bitta bo'lakka necha kvadrat metr maysa ekiladi? (π = 3,14)**

A) 803,84 m² **B)** 401,92 m² **C)** 12,56 m² **D)** 100,48 m²

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11. **Radiusi 5 cm bo'lgan 4 ta bir xil doira rasmdagidek bir-biriga tegib turibdi. Ular orasidagi bo'yalgan soha yuzini toping (π = 3,14).**



A) 43 cm² **B)** 21,5 cm²
C) 100 cm² **D)** 78,5 cm²

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12. **Ikki bola bir-biriga qarab bir xil chambarak dumalatdi va chambaraklar 12 sekunddan keyin to'qnashdi. Chambarakning radiusi 21 cm, har biri 2 sekundda bir marta to'liq aylanadi. Boshida chambaraklar orasida necha metr bo'lgan? (π = $\frac{22}{7}$)**

A) 7,92 m **B)** 2,64 m **C)** 31,68 m **D)** 15,84 m

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13. **Ikki dumaloq maysazor bor: kattasining radiusi kichiginikidan 2 marta katta. Kichik maysazorga 3 kg o'g'it sepildi. Katta maysazorning har 1 m² iga ham xuddi shuncha o'g'it sepilsa, unga necha kg o'g'it kerak?**

A) 12 kg **B)** 24 kg **C)** 9 kg **D)** 6 kg

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14. **Katta yarim doiraning diametri 30 cm. Uning ichidan diametrlari 18 cm va 12 cm bo'lgan ikkita yarim doira kesib olindi (diametrlari katta diametr ustida). Qolgan shaklning perimetrini toping (π = 3,14).**

A) 94,2 cm **B)** 141,3 cm **C)** 47,1 cm **D)** 124,2 cm

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15. **Tomoni 30 cm bo'lgan kvadrat tunukadan diametri 10 cm bo'lgan doiralar kesib olindi: 3 qator, har qatorda 3 tadan. Ortib qolgan tunuka yuzini toping (π = 3,14).**

A) 586 cm² **B)** 821,5 cm² **C)** 617,4 cm² **D)** 193,5 cm²

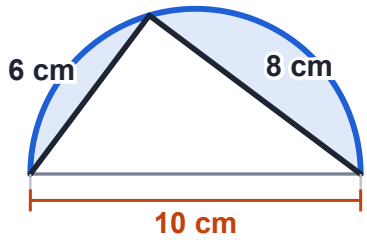
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16. **Yarim doiraning diametri 12 cm. Diametr ustidan, yarim doira ichiga, diametri 6 cm bo'lgan ikkita yarim doira o'yib olindi. Qolgan shaklning perimetrini π orqali ifodalang.**

A) (12π + 12) cm
B) (6π + 12) cm
C) 6π cm
D) 12π cm

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17. **Yarim doiraning diametri 10 cm. Uning ichida tomonlari 6 cm, 8 cm va 10 cm bo'lgan to'g'ri burchakli uchburchak bor (rasm). Bo'yalgan soha yuzini toping (π = 3,14).**



A) 15,25 cm²
B) 63,25 cm²
C) 54,5 cm²
D) 39,25 cm²

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18. **Gulzor to'g'ri to'rtburchak (8 m × 5 m) va uning uzun tomoniga chizilgan yarim doiradan iborat. 1 m² ga maysa ekish 20 000 so'm turadi. Butun gulzorga maysa ekish necha so'm turadi? (π = 3,14)**

A) 1 804 800 so'm
B) 2 809 600 so'm
C) 800 000 so'm
D) 1 302 400 so'm

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19. **Kvadrat ichiga eng katta doira chizildi. Kvadratning doiradan tashqaridagi to'rt burchagidagi qismlar birgalikda 3,44 cm². Kvadratning tomoni necha cm? (π = 3,14)**

A) 8 cm
B) 4 cm
C) 16 cm
D) 2 cm

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20. **Shakl kvadrat (tomoni 10 cm) va uning bir tomoniga tashqaridan chizilgan yarim doiradan iborat. Sardor shakl perimetrini 55,7 cm deb topdi (π = 3,14). Uning xatosi nimada?**

A) Ichkaridagi tomonni ham qo'shgan
B) Yoy uzunligini noto'g'ri topgan
C) Diametr o'rniga radius olgan
D) Xato qilmagan

Javoblar

Murakkab shakllarning yuzi va perimetri · 2-test (qiyinroq)

№	Javob	Qiymati
1	**C**	43 cm²
2	**B**	51,4 cm
3	**B**	165,94 m²
4	**C**	94,2 m
5	**A**	210 cm²
6	**A**	(32π − 64) cm²
7	**A**	41,12 cm
8	**A**	64 cm²
9	**D**	44,56 m
10	**D**	100,48 m²

№	Javob	Qiymati
11	**B**	21,5 cm²
12	**D**	15,84 m
13	**A**	12 kg
14	**A**	94,2 cm
15	**D**	193,5 cm²
16	**D**	12π cm
17	**A**	15,25 cm²
18	**D**	1 302 400 so'm
19	**B**	4 cm
20	**A**	Ichkaridagi tomonni ham qo'shgan

Qisqa yechimlar

1. To'g'ri to'rtburchak: 20 · 10 = 200 cm².

Ikkita chorak doira: $2 \cdot \frac{1}{4} \cdot 3{,}14 \cdot 10 \cdot 10 = 157$ cm².

Bo'yalgan soha: 200 − 157 = **43 cm²**.

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2. Chegarada: tepadagi tomon 20 cm va ikkita chorak yoy.

Bitta yoy: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 10 = 15{,}7$ cm.

Perimetr: 20 + 15,7 + 15,7 = **51,4 cm**.

(71,4 — chorak doiralar ichidagi yon tomonlar ham qo'shilgan.)

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3. Ikki uzun tomon bo'ylab to'g'ri bo'laklar: 2 · 50 · 1 = 100 m².

Ikki chetdagi yarim halqalar — bitta halqa: ichki radius 20 : 2 = 10 m, tashqi 10 + 1 = 11 m.

3,14 · 11 · 11 − 3,14 · 10 · 10 = 379,94 − 314 = 65,94 m².

Qoplama: 100 + 65,94 = **165,94 m²**.

(231,88 — har bir yarim halqa butun halqa deb olingan.)

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4. Shakl chegarasida faqat 3 ta yarim aylana yoyi bor, uchburchak tomonlari ichkarida.

$\frac{1}{2} \cdot 3{,}14 \cdot 15 + \frac{1}{2} \cdot 3{,}14 \cdot 20 + \frac{1}{2} \cdot 3{,}14 \cdot 25 = \frac{1}{2} \cdot 3{,}14 \cdot 60$ = **94,2 m**.

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5. Kichik yarim doiralar: $\frac{1}{2} \cdot 3{,}14 \cdot (10 \cdot 10 + 10{,}5 \cdot 10{,}5) = 330{,}0925$ cm².

Katta yarim doira: $\frac{1}{2} \cdot 3{,}14 \cdot 14{,}5 \cdot 14{,}5 = 330{,}0925$ cm² — xuddi shuncha!

Bo'yalgan soha = kichik yarim doiralar + uchburchak − katta yarim doira = uchburchak yuzi: $\frac{1}{2} \cdot 20 \cdot 21$ = **210 cm²**.

(540,0925 — katta yarim doira ayirilmagan.)

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6. Bitta chorak doira: $\frac{1}{4} \cdot \pi \cdot 8 \cdot 8 = 16\pi$ cm². Ikkitasi: 32π cm².

Ikkala chorak doirada yaproq ikki marta, qolgan qism bir marta sanaldi — kvadrat yuzini ayiramiz: 8 · 8 = 64 cm².

Yaproq: **(32π − 64) cm²**.

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7. Yon tomonlar: 8 + 8 = 16 cm.

Ikkita yarim aylana yoyi: $2 \cdot \frac{1}{2} \cdot 3{,}14 \cdot 8 = 25{,}12$ cm.

Perimetr: 16 + 25,12 = **41,12 cm**.

(57,12 — punktir chiziqlar ham qo'shilgan.)

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8. Kesib olingan yarim doira yo'qolmadi — pastga ko'chdi.
Shuning uchun yuz kvadratnikidek: 8 · 8 = **64 cm²**.

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9. Bo'lak — doiraning $\frac{1}{8}$ qismi.
Yoy: $\frac{1}{8}$ · 2 · 3,14 · 16 = 12,56 m.
Ikkita radius: 16 + 16 = 32 m.
To'siq: 12,56 + 32 = **44,56 m**.

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10. Radius: 32 : 2 = 16 m.
Butun gulzor: 3,14 · 16 · 16 = 803,84 m².
Bitta bo'lak: 803,84 : 8 = **100,48 m²**.
(401,92 — radius o'rniga diametr olingan.)

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11. Markazlarni tutashtirsak, tomoni 5 + 5 = 10 cm bo'lgan kvadrat chiqadi: 10 · 10 = 100 cm².
Kvadrat ichida har bir doiraning choragi bor — 4 ta chorak, ya'ni bitta doira: 3,14 · 5 · 5 = 78,5 cm².
Bo'yalgan soha: 100 − 78,5 = **21,5 cm²**.

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12. Har bir chambarak 12 : 2 = 6 marta aylandi.
Bir aylanishda chambarak aylana uzunligicha yuradi: 2 · $\frac{22}{7}$ · 21 = 132 cm.
Bittasi: 6 · 132 = 792 cm. Ikkalasi: 2 · 792 = 1584 cm = **15,84 m**.
(7,92 — faqat bitta chambarakning yo'li.)

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13. Yuz = π · r · r — radius ikki marta ko'paytiriladi.
Radius 2 marta katta bo'lsa, yuz 2 · 2 = 4 marta katta.
O'g'it ham 4 marta ko'p: 3 · 4 = **12 kg**.
(6 kg — yuz ham 2 marta katta deb o'ylansa.)

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14. Chegarada 3 ta yarim aylana yoyi bor, to'g'ri chiziq yo'q.
Katta yoy: $\frac{1}{2}$ · 3,14 · 30 = 47,1 cm.
Kichik yoylar: $\frac{1}{2}$ · 3,14 · 18 + $\frac{1}{2}$ · 3,14 · 12 = 28,26 + 18,84 = 47,1 cm.
Perimetr: 47,1 + 47,1 = **94,2 cm**.

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15. Doiralar soni: 3 · 3 = 9 ta, har birining radiusi 5 cm.
Bitta doira: 3,14 · 5 · 5 = 78,5 cm²; 9 tasi: 706,5 cm².
Kvadrat: 30 · 30 = 900 cm².
Ortib qoldi: 900 − 706,5 = **193,5 cm²**.
(617,4 — doira yuzi o'rniga aylana uzunligi olingan.)

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16. Katta yoy: $\frac{1}{2}$ · π · 12 = 6π cm.
Ikkita kichik yoy: 2 · $\frac{1}{2}$ · π · 6 = 6π cm.
To'g'ri chiziq chegarada qolmadi. Perimetr: 6π + 6π = **12π cm**.

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17. Yarim doiraning radiusi 5 cm: $\frac{1}{2}$ · 3,14 · 5 · 5 = 39,25 cm².
Uchburchak (6 va 8 — to'g'ri burchak tomonlari): $\frac{1}{2}$ · 6 · 8 = 24 cm².
Bo'yalgan soha: 39,25 − 24 = **15,25 cm²**.

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18. Yarim doiraning radiusi 8 : 2 = 4 m: $\frac{1}{2}$ · 3,14 · 4 · 4 = 25,12 m².
Gulzor: 8 · 5 + 25,12 = 65,12 m².
Narxi: 65,12 · 20 000 = **1 302 400 so'm**.

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19. Variantlarni tekshiramiz. Doiraning diametri kvadrat tomoniga teng.
Tomoni 4 cm bo'lsa, radius 2 cm: 4 · 4 − 3,14 · 2 · 2 = 16 − 12,56 = 3,44 cm² ✓
Tomoni 2 cm bo'lsa: 4 − 3,14 = 0,86 cm²; tomoni 8 cm bo'lsa: 64 − 50,24 = 13,76 cm² — mos emas.
Javob: **4 cm**.

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20. To'g'ri hisob: 3 ta tomon 10 + 10 + 10 = 30 cm, yoy $\frac{1}{2} \cdot 3{,}14 \cdot 10 = 15{,}7$ cm; jami 45,7 cm.
55,7 − 45,7 = 10 cm — bu kvadratning yarim doira bilan tutashgan tomoni.
Sardor **ichkaridagi tomonni ham qo'shgan**.
